THE NEUTRALITY THEOREM

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Abstract. The centered collision fluctuation sum $F^\circ(s)$ converges at $s = 1$ and appears to converge below $s = 1$ into the critical strip. This paper identifies the mechanism.

The centered sum decomposes as $F^\circ = F^{\circ\circ} + \mathcal{M}$, where $F^{\circ\circ}$ is doubly centered and $\mathcal{M}$ is the mod-3 component. In the nontrivial cancellation regime $c_0 \neq 0$, both diverge individually below $s = 1$, but their ratio approaches -1 with increasing precision. At finite range the deviation $(1 + \text{ratio}) \sim C/|F^{\circ\circ}|^\alpha$ fits $\alpha \approx 0.6$, but the character decomposition proves $\alpha = 1$: the cancellation is perfect.

The proof is algebraic. The mod-3 component decomposes over characters modulo 3 as $\mathcal{M} = c_0 \sum_p p^{-s} + c_1 \sum_p \chi_3(p)\, p^{-s}$. The doubly centered sum carries the same Mertens term $-c_0 \sum_p p^{-s}$ with opposite sign. In $F^\circ = F^{\circ\circ} + \mathcal{M}$, the Mertens terms cancel exactly, leaving F° as a pure linear combination of non-trivial character sums. These converge wherever the corresponding L-functions have no zeros.

1. Introduction

This paper studies why the centered collision fluctuation sum $F^\circ(s)$ penetrates below $s = 1$.

The preceding papers [6, 7] proved convergence at $s = 1$ via the collision transform and Mertens' theorem, and observed computational convergence to at least $s = 0.6$ in base 10. The mod-3 polarity was identified as the structural feature controlling the penetration depth: removing the mod-3 component preserves convergence at $s = 1$ but destroys it below.

The present paper proves that the mod-3 cancellation is algebraically perfect. The doubly centered sum $F^{\circ\circ}$ and the mod-3 component $\mathcal{M}$ share a common Mertens divergence $\pm c_0 \sum_p p^{-s}$ that cancels in their sum. The residual F° contains only non-trivial character sums.

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2. The Decomposition

Definition 1. Let $S^\circ(a)$ denote the singly centered collision deviation (centered by spectral class). For each unit $a \bmod m$ with $m = b^{\ell+1}$, define:

$$\mu_3(a) = \frac{1}{|\{a' : a' \equiv a \pmod 3\}|} \sum_{a' \equiv a \pmod 3} S^\circ(a'),$$

$$S^{\circ\circ}(a) = S^\circ(a) - \mu_3(a).$$

The *mod-3 component* is $\mathcal{M}(s) = \sum_p \mu_3(p)/p^s$ and the *doubly centered sum* is $F^{\circ\circ}(s) = \sum_p S^{\circ\circ}(p)/p^s$.

By construction:

$$F^\circ(s) = F^{\circ\circ}(s) + \mathcal{M}(s).$$

Proposition 2. *The antisymmetry $S^\circ(a) = -S^\circ(m-a)$ is preserved by the mod-3 centering: $S^{\circ\circ}(a) = -S^{\circ\circ}(m-a)$.*

Proof. The reflection $a \mapsto m - a$ pairs mod-3 classes: it sends residue class r to $m - r \pmod 3$. Since S° is antisymmetric, the mod-3 means of reflected classes satisfy

$$\mu_3(r) = -\mu_3(m - r).$$

Therefore

$$S^{\circ\circ}(m - a) = S^\circ(m - a) - \mu_3(m - a) = -S^\circ(a) + \mu_3(a) = -S^{\circ\circ}(a).$$

□

Corollary 3. *$F^{\circ\circ}(1)$ converges: the antisymmetry ensures even characters vanish, and the proof of the Centered Collision PNT* [6] *applies.*

3. The Character Algebra

The function $\mu_3(a)$ depends only on $a \bmod 3$. When $3 \nmid b$, we have $\gcd(3, m) = 1$, so the restriction to primes $p > 3$ (where $p \bmod 3 \in \{1, 2\}$) gives a decomposition over characters modulo 3:

$$\mu_3(p) = c_0 + c_1\chi_3(p),$$

where χ_3 is the non-trivial character modulo 3,

$$c_0 = \frac{m_1 + m_2}{2}, \qquad c_1 = \frac{m_1 - m_2}{2},$$

and m_1, m_2 are the mod-3 means: $m_j = \mu_3(a)$ for any $a \equiv j \pmod 3$.

In base 10 at lag 1: $m_1 = -1.30$, $m_2 = 0$, giving $c_0 = c_1 = -0.65$.

The mod-3 component is therefore

$$\mathcal{M}(s) = c_0 \sum_p \frac{1}{p^s} + c_1 \sum_p \frac{\chi_3(p)}{p^s}. \tag{1}$$

The first sum is the Mertens prime sum, diverging like $\log\log x$ at $s = 1$ and like $x^{1-s}/((1-s)\log x)$ for $s < 1$. The second converges at $s = 1$ by Dirichlet's theorem [1].

4. The Cancellation Theorem

Theorem 4 (Perfect cancellation)**.** *The Mertens divergence in $F^{\circ\circ}$ and $\mathcal{M}$ cancels exactly:*

$$F^{\circ}(s) = \sum_{\substack{\chi \bmod m \\ \chi(-1)=-1}} \hat{S}^{\circ}(\chi)\, P(s,\chi),$$

where $P(s,\chi) = \sum_p \chi(p)/p^s$. No trivial character (Mertens) term appears.

Proof. By the collision transform [6], $S^{\circ}(p) = \sum_\chi \hat{S}^{\circ}(\chi)\,\chi(p)$, and the antisymmetry theorem restricts the sum to odd characters ($\chi(-1) = -1$). Therefore

$$F^{\circ}(s) = \sum_{\substack{\chi \bmod m \\ \chi(-1)=-1}} \hat{S}^{\circ}(\chi) \sum_p \frac{\chi(p)}{p^s} = \sum_{\substack{\chi \bmod m \\ \chi(-1)=-1}} \hat{S}^{\circ}(\chi)\, P(s,\chi).$$

Every character in this sum is non-trivial (the trivial character is even), so no Mertens term $\sum_p p^{-s}$ appears.

The decomposition $F^{\circ} = F^{\circ\circ} + \mathcal{M}$ introduces the Mertens term $c_0 \sum_p p^{-s}$ via (1), and $F^{\circ\circ}$ must carry $-c_0 \sum_p p^{-s}$ to compensate. In F°, these cancel algebraically. □

Definition 5. The *cancellation exponent* α is defined by

$$1 + \frac{F^{\circ\circ}(s)}{\mathcal{M}(s)} \sim \frac{C}{|F^{\circ\circ}(s)|^{\alpha}}$$

as $|F^{\circ\circ}(s)| \to \infty$.

Corollary 6. *Assume $c_0 \neq 0$, so that the Mertens term in* (1) *is present and $|F^{\circ\circ}(s)| \to \infty$. Then the cancellation exponent is $\alpha = 1$.*

Proof. Since $1 + F^{\circ\circ}/\mathcal{M} = F^{\circ}/\mathcal{M}$, and $F^{\circ}(s)$ converges to a finite value while $|\mathcal{M}(s)| \sim |c_0| \sum_p p^{-s} \to \infty$, the ratio $F^{\circ}/\mathcal{M} \to 0$. Since

$F^{\circ\circ} = F^{\circ} - \mathcal{M}$ and $F^{\circ}(s)$ converges while $|\mathcal{M}(s)| \to \infty$, we have $F^{\circ\circ}/\mathcal{M} = F^{\circ}/\mathcal{M} - 1 \to -1$, so $|F^{\circ\circ}| \sim |\mathcal{M}|$. Therefore

$$\big|1 + \text{ratio}\big| = \frac{|F^{\circ}(s)|}{|\mathcal{M}(s)|} \sim \frac{C}{|F^{\circ\circ}(s)|^{1}}.$$

□

5. Computational Verification

primes	s	F°	$F^{\circ\circ}$	$\mathcal{M}$	1+ratio	α_{fit}
10,000	0.7	+0.27	+7.1	−6.8	−0.039	0.58
100,000	0.7	+0.28	+13.1	−12.8	−0.022	0.61
348,488	0.7	+0.29	+18.1	−17.8	−0.016	0.62
700,000	0.7	+0.26	+21.6	−21.3	−0.012	0.91
1,270,000	0.7	+0.27	+25.2	−24.9	−0.011	0.75

Table 1. The fitted cancellation exponent α drifts upward as the prime range increases. At 348,488 primes $\alpha \approx 0.6$; at 1,270,000 primes the fit is noisy but trends toward $\alpha = 1$. Base 10, lag 1.

The finite-range exponent $\alpha \approx 0.6$ reported in earlier computation reflects the pre-asymptotic regime. As the prime count increases, the Mertens term $c_0 \sum_p p^{-s}$ dominates both $F^{\circ\circ}$ and $\mathcal{M}$, and the ratio $F^{\circ}/\mathcal{M}$ shrinks as $1/|\mathcal{M}|$, confirming $\alpha = 1$.

s	F°	$\lvert F^{\circ\circ}\rvert$	(1+ratio) $\cdot$ $\lvert F^{\circ\circ}\rvert$
1.0	+0.08	0.90	0.085
0.9	+0.12	2.38	0.126
0.8	+0.18	7.25	0.187
0.7	+0.27	25.2	0.273
0.6	+0.37	96.6	0.369
0.5	+0.37	399.6	0.370

Table 2. With $\alpha = 1$, the product (1+ratio) $\cdot$ $|F^{\circ\circ}|$ equals $|F^{\circ}(s)|$, which is bounded. Compare with Table 1: the product stabilizes as the prime count grows. 1,270,000 primes.

Three regimes emerge. When $3 \mid b$, the modulus $m \equiv 0 \pmod 3$ forces all units into mod-3 classes 1 and 2, and the centering removes

b	$\phi(b^2)$	c_0	$F^\circ(0.5)$	$\lvert F^{\circ\circ}(0.5)\rvert$	α_{fit}	type
3	6	0	+0.65	0.65	—	$3 \mid b$
5	20	+0.11	+0.88	51	0.69	weak
7	42	−0.70	+1.76	323	0.69	strong
10	40	−0.65	+0.09	296	0.93	strong
11	110	+0.02	+1.72	8.3	—	weak
12	48	0	−0.64	0.64	—	$3 \mid b$
13	156	−0.80	+3.53	369	0.60	strong

TABLE 3. The cancellation exponent across bases (664,000 primes). When $3 \mid b$, the mod-3 bias vanishes ($c_0 = 0$) and $F^\circ = F^{\circ\circ}$ with no cancellation needed. When $3 \nmid b$, the bias is strong ($|c_0| > 0.5$) or weak ($|c_0| < 0.2$). At strong bias, α is already near 1 at this range. At weak bias, the Mertens term has not yet dominated and the finite-range α is lower.

the bias completely: $c_0 = 0$ and no cancellation mechanism is needed. When $3 \nmid b$ and the bias is strong ($|c_0| \approx 0.7$), the Mertens term dominates early and α approaches 1 quickly. When $3 \nmid b$ but the bias is weak ($|c_0| \approx 0.02$), the character sum corrections remain comparable to the Mertens term at this range, and the finite-range α stays low. In the nontrivial cancellation regime $c_0 \neq 0$, the theorem predicts $\alpha = 1$ asymptotically; when $c_0 = 0$, no cancellation exponent is defined because no divergent Mertens term is present.

6. Consequences

Corollary 7. *$F^\circ(s)$ converges for $\mathrm{Re}(s) > 1 - c/\log m$, where $c > 0$ is the constant from the classical zero-free region for Dirichlet L-functions modulo m.*

Proof. By Theorem 4, $F^\circ(s) = \sum \hat{S}^\circ(\chi)\, P(s,\chi)$ over non-trivial odd characters. By partial summation and the explicit formula for $\sum_{p \le x} \chi(p)$ (see [1, Ch. 20] or [4, Ch. 13]), each $P(s,\chi)$ converges wherever $L(s,\chi)$ has no zeros with real part $\ge \mathrm{Re}(s)$. The classical zero-free region [1, 4] guarantees this for $\mathrm{Re}(s) > 1 - c/\log m$. □

Remark 8. By the convergence theorem for Dirichlet series [2], convergence of F° at a real point s_0 extends to the half-plane $\mathrm{Re}(s) > s_0$. The question of penetration depth is therefore a question about the real abscissa of convergence.

Computation indicates convergence to at least $s_0 = 0.5$ [7]. If confirmed, this would imply convergence for all $\mathrm{Re}(s) > 1/2$. Since F° is a finite linear combination of character sums $P(s, \chi)$, two scenarios are possible.

If the products $\hat{S}^\circ(\chi) \cdot P(s, \chi)$ were to have non-negative real part, then convergence of F° requires each $P(s, \chi)$ to converge individually, which is equivalent to $L(s, \chi) \neq 0$ for $\mathrm{Re}(s) > 1/2$: the Generalized Riemann Hypothesis for the odd L-functions modulo m.

If the positivity condition fails, convergence of F° could arise from cancellation between divergent character sums. This would mean the collision coefficients $\hat{S}^\circ(\chi)$ encode structural information about the zeros of the corresponding L-functions.

7. The Neutrality Theorem

The mod-3 component $\mathcal{M}$ arises because the collision invariant S has unequal means across mod-3 classes. The following theorem identifies the universal structure.

Theorem 9 (Neutrality of the fixed class)**.** *For any base b with $3 \nmid b$ and any lag $\ell \geq 1$, let $m = b^{\ell+1}$ and let $r_m \in \{1, 2\}$ be the unique residue class satisfying*

$$2r_m \equiv m \pmod 3 \qquad \textit{equivalently} \qquad r_m \equiv -m \pmod 3.$$

Then the mean of S (the collision deviation at lag ℓ) over units $a \equiv r_m \pmod 3$ in $(\mathbb{Z}/b^{\ell+1}\mathbb{Z})^\times$ equals $-1/2$ (the grand mean). In particular, when $m \equiv 1 \pmod 3$ the neutral class is 2, and when $m \equiv 2 \pmod 3$ the neutral class is 1.

Proof. Since $3 \nmid b$, the class r_m is well-defined modulo 3. The reflection $a \mapsto m - a$ sends residue class r to $m - r \pmod 3$. The class r_m is fixed by this map because

$$m - r_m \equiv r_m \pmod 3 \qquad \Longleftrightarrow \qquad 2r_m \equiv m \pmod 3.$$

Thus reflection is a self-map on the class-r_m units. The reflection identity $S(a) + S(m-a) = -1$ pairs units within this class, and each pair averages to $-1/2$. □

Corollary 10. *The remaining two mod-3 classes are swapped by reflection, so their means sum to -1. Thus all deviation from the grand mean is concentrated in the reflected pair, with equal magnitude and opposite sign around $-1/2$.*

For the lag-1 case studied in the critical-strip note, $m = b^2 \equiv 1 \pmod 3$ whenever $3 \nmid b$, so the neutral class is 2 and the reflected pair is

$\{0, 1\}$. Since primes $p > 3$ satisfy $p \bmod 3 \in \{1, 2\}$, the prime harmonic sum then sees only classes 1 (which carries a base-dependent bias) and 2 (which is neutral). This is the source of the lag-1 mod-3 polarity used in the critical-strip computations.

Remark 11. Since $3 \mid 3$, Theorem 9 does not apply to base 3. In base 3, the class-1 mean is $-2/3$ and the class-2 mean is $-1/3$, giving a splitting ratio of $|-2/3|/(|-2/3|+|-1/3|) = 2/3$. The fraction $2/3 = (p-1)/p$ at $p = 3$ is the alignment limit of the prime 3 [5]. This ratio is specific to base 3; in the bases with $3 \nmid b$ and $m \equiv 1 \pmod 3$, the class-2 mean is $-1/2$ (neutral), while in the cases with $m \equiv 2 \pmod 3$ the neutral class is 1. The universal structure is the neutrality of the fixed reflected class, not the ratio 2/3.

Computation confirms the neutrality theorem across all bases $b \leq 40$ with $3 \nmid b$: in the cases with $m \equiv 1 \pmod 3$ one finds $2 \cdot \mathrm{sum}_2 = -\mathrm{count}_2$ exactly, while in the cases with $m \equiv 2 \pmod 3$ the analogous identity holds with class 1 in place of class 2. For bases with $3 \mid b$, the modulus $m \equiv 0 \pmod 3$, no units satisfy $a \equiv 0 \pmod 3$, and the structure is different.

8. Remarks

What is proved. The decomposition $F^\circ = F^{\circ\circ} + \mathcal{M}$ is exact. In the nontrivial cancellation regime $c_0 \neq 0$, the character algebra proves $\alpha = 1$: the Mertens divergence cancels algebraically, leaving F° as a pure non-trivial character sum. When $c_0 = 0$, no cancellation is needed. Convergence extends to the classical zero-free region for L-functions modulo m [1]. The neutrality of the fixed mod-3 reflected class is proved from the reflection identity.

What the computation shows. At finite range (348,488 primes), the fitted exponent $\alpha \approx 0.6$ reflects the pre-asymptotic regime where character sum corrections are comparable to the Mertens term. As the range increases to 1,270,000 primes, the fit drifts toward $\alpha = 1$, confirming the theorem.

The arc. The alignment limit 2/3 from [5] is the base-3 instance of the neutrality theorem: at base 3, the class-1 bias is $-2/3$ and the class-2 mean is $-1/3$ (neutral at $-1/2$ in the centered picture). The universal structure is not the ratio 2/3 but the neutrality of the fixed reflected class, which follows from the bilateral symmetry of the digit function. The collision transform [6] provides the character decomposition, and the antisymmetry theorem [7] restricts to odd characters, eliminating the Mertens term and producing perfect cancellation.

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